

b.) Existential quantifiers

Example Consider the following statement:

There exist 3×3 matrices A and B such that $AB = BA$.

We can write this statement as follows:

$$\exists A \in M_{3 \times 3}(\mathbb{R}) \exists B \in M_{3 \times 3}(\mathbb{R}) \quad AB = BA.$$

This is a true statement.

To prove that such a statement is true, it is sufficient to give one example of A and B where $AB = BA$ is true. We can for instance take

$$A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Example Consider the statement

$$\exists x \in \mathbb{Q} \text{ s.t. } x^2 = 2$$

False statement

You have to prove the non-existence of something. This is often harder.

existential quantifier

Quantifiers and implication

Example For all $n \times n$ matrices A and B , if $\underbrace{AB \neq 0}_P$ then $\underbrace{A \neq 0 \text{ and } B \neq 0}_Q$.

P

Q

$$\forall A, B \in M_{n \times n}(\mathbb{R}) \quad \underbrace{AB \neq 0}_P \implies \underbrace{A \neq 0 \wedge B \neq 0}_Q \quad (*)$$

true statement

Converse of statement (*):

$$\forall A, B \in M_{n \times n}(\mathbb{R}) \quad \underbrace{A \neq 0 \wedge B \neq 0}_Q \implies \underbrace{AB \neq 0}_P$$

false statement

(Take for instance $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$.)

The converse is not the same as the original statement!

The contrapositive

The contrapositive of the statement $P \implies Q$ is the statement $\neg Q \implies \neg P$.

The contrapositive of a statement says exactly the same as the statement (either both are true or both are false).

Sometimes the contrapositive is easier to prove.

Example

Contrapositive of (*):

$$\forall A, B \in M_{n \times n}(\mathbb{R}), \quad A = 0 \vee B = 0 \implies AB = 0$$

true statement

Remark: Universal quantifiers are often implicit.

Example:

Theorem If A and B are invertible matrices of the same size then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

This needs to be read as

universal quantification



Theorem For all matrices A and B , if A and B are invertible matrices of the same size then AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$.

Elementary matrices and invertibility

see Sect 1.6 of Notes

Notation for elementary row operations

Example

e = "multiply row 3 with 2" = " $R_3 \rightarrow 2R_3$ "

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$.

$e(A)$ is always a matrix!

Then $e(A) = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 14 & 16 & 18 \end{pmatrix}$

e maps matrices to matrices. It is a function.

e^{-1} is the inverse elementary operation of e . For instance

$$e^{-1} \left(\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 14 & 16 & 18 \end{pmatrix} \right) = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

Elementary matrices

Definition An elementary matrix is a matrix obtained by performing one (and only one) row operation on an identity matrix, i.e., a matrix $e(I_n)$, where e is an elementary row operation.

3 types of elementary matrix

a.) Add a multiple of a row to another row

Ex. Let $e_1 = "R_2 \rightarrow R_2 - 4R_1"$.

Then $e_1 \left(\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \right) = \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{pmatrix}$

Elementary matrix corresponding to e_1 :

$$e_1 \left(\underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{I_3} \right) = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{E_1}$$

It holds that

$$E_1 \cdot A = e_1(A)$$

$$\underbrace{\begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{E_1} \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}}_A = \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 7 & 8 & 9 \end{pmatrix}}_{e_1(A)}$$

Multiplying by E_1 on the left has the same effect as applying the elementary operation e_1 .

What is the inverse of E_1 ?

Recall

$$e_1 = "R_2 \rightarrow R_2 - 4R_1"$$

Inverse operation :

$$e_1^{-1} = "R_2 \rightarrow R_2 + 4R_1"$$

Hence

$$E_1^{-1} = e^{-1}(I_3) = \begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

You may check that

$$\begin{pmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -4 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(and also the other way round)

Theorem If $E = e(I)$ then $E^{-1} = e^{-1}(I)$.

Proof see Notes.

b.) Multiply a row by a non-zero number

Ex $e_2 = "R_2 \rightarrow 5R_2"$

$$e_2 \left(\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \right) = \begin{pmatrix} 1 & 2 \\ 15 & 20 \end{pmatrix}$$

Elementary matrix corresponding to e_2 (for 2×2 matrices):

$$e_2(I_2) = e_2 \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) = \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix} \quad E_2$$

$$\begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 15 & 20 \end{pmatrix} \quad E_2$$

$$e_2^{-1} = "R_2 \rightarrow \frac{1}{5}R_2"$$

Hence

$$E_2^{-1} = e_2^{-1}(I_2) = e_2^{-1} \left(\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right) = \begin{pmatrix} 1 & 0 \\ 0 & 1/5 \end{pmatrix}$$

c.) Swapping two rows

Ex. $e_3 = "R_1 \leftrightarrow R_2"$

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

$$e_3(A) = \begin{pmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \end{pmatrix}$$

Elementary matrix corresponding to e_3 (for 2×3 matrices):

$$E_3 = e_3(I_2) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

We can verify that

$$E_3 A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} = \begin{pmatrix} 4 & 5 & 6 \\ 1 & 2 & 3 \end{pmatrix}, \quad e_3(A)$$

as desired.

Note that

$$E_3^{-1} = e_3^{-1}(I_2) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad E_3$$

An elementary matrix corresponding to the swapping of two rows is always self-inverse.